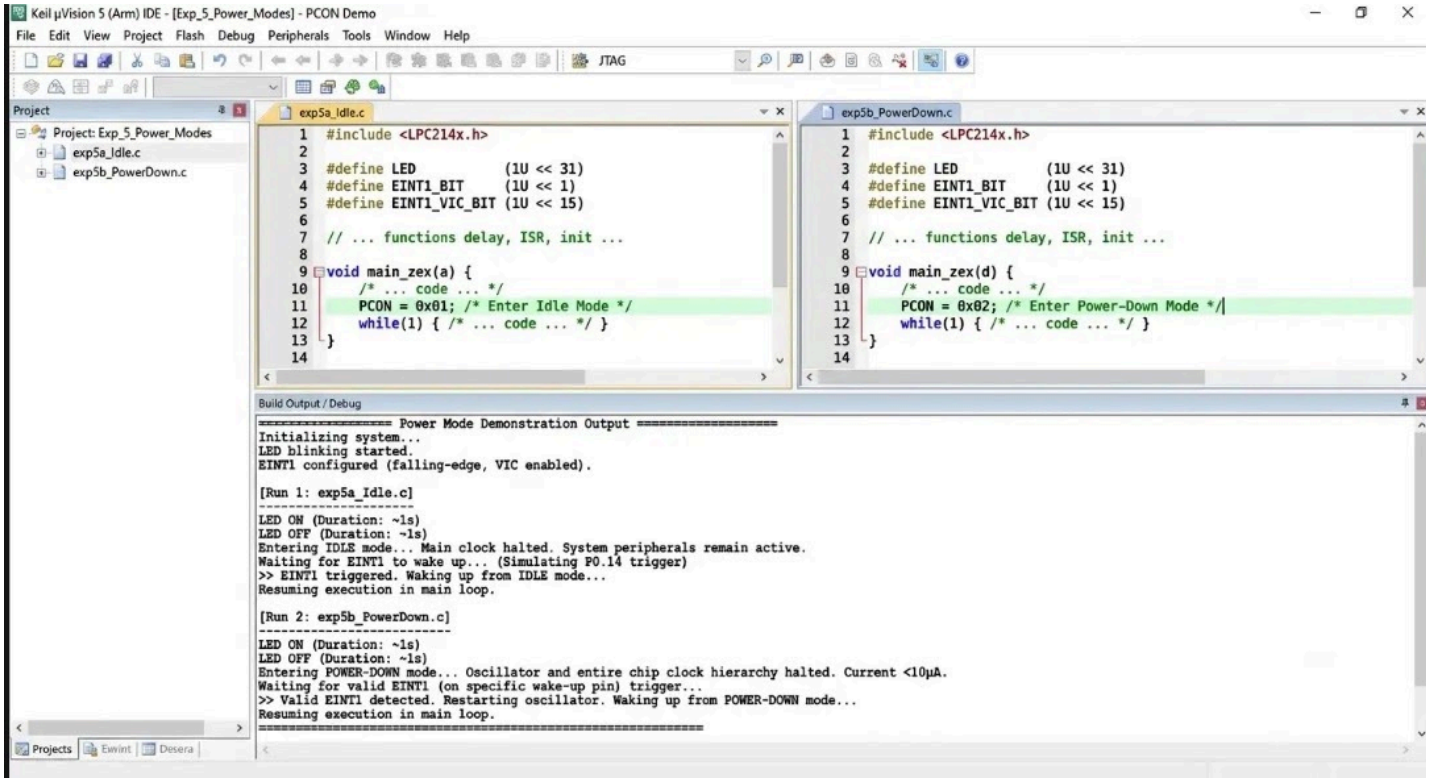

Keil µVision 5 (Arm) IDE - [Exp_5_Power_Modes] - PCON Demo
File Edit View Project Flash Debug Peripherals Tools Window Help
JTAG
Project
Project: Exp_5_Power_Modes
exp5a_Idle.c
exp5b_PowerDown.c
exp5a_Idle.c
1 #include <LPC214x.h>
2
3 #define LED (1U << 31)
4 #define EINT1_BIT (1U << 1)
5 #define EINT1_VIC_BIT (1U << 15)
6
7 // ... functions delay, ISR, init ...
8
9 void main_zex(a) {
10 /* ... code ... */
11 PCON = 0x01; /* Enter Idle Mode */
12 while(1) { /* ... code ... */ }
13 }
14
exp5b_PowerDown.c
1 #include <LPC214x.h>
2
3 #define LED (1U << 31)
4 #define EINT1_BIT (1U << 1)
5 #define EINT1_VIC_BIT (1U << 15)
6
7 // ... functions delay, ISR, init ...
8
9 void main_zex(d) {
10 /* ... code ... */
11 PCON = 0x02; /* Enter Power-Down Mode */
12 while(1) { /* ... code ... */ }
13 }
14
Build Output / Debug
========== Power Mode Demonstration Output ==========
Initializing system...
LED blinking started.
EINT1 configured (falling-edge, VIC enabled).
[Run 1: exp5a_Idle.c]
LED ON (Duration: ~1s)
LED OFF (Duration: ~1s)
Entering IDLE mode... Main clock halted. System peripherals remain active.
Waiting for EINT1 to wake up... (Simulating P0.14 trigger)
>> EINT1 triggered. Waking up from IDLE mode...
Resuming execution in main loop.
[Run 2: exp5b_PowerDown.c]
LED ON (Duration: ~1s)
LED OFF (Duration: ~1s)
Entering POWER-DOWN mode... Oscillator and entire chip clock hierarchy halted. Current <10µA.
Waiting for valid EINT1 (on specific wake-up pin) trigger...
>> Valid EINT1 detected. Restarting oscillator. Waking up from POWER-DOWN mode...
Resuming execution in main loop.
Projects
Ewint
Desera